

hepatotoxic and nephro-toxic effects. Inhalation of airborne copper causes metal fume fever and irritation of the respiratory tract.

Lead. Lead (Pb) is the most ancient industrial poison known to man. Lead poisoning is also known as plumbism or saturnism. The earth contains about 0.00002 per cent of lead by weight. Lead is the last member of the carbon family and exhibits metallic properties. It has weak electropositive character and electro-acceptor property.

Lead is a significant material in CRTs, accounting for up to 8 per cent of overall composition of the CRT by weight. It can build up in the body through repeated exposure and have irreversible effects on the nervous and endocrine systems. It also has serious negative effects on children's brain development.

Manganese. Manganese (Mn) is essential for the well being of cells and functions as a co-factor in a number of enzymatic reactions, particularly those involved in phosphorylation, cholesterol and fatty acids synthesis. It is a toxic at higher concentrations and is used in the manufacture of dry battery cells, electrical coils and cables, and ceramic wares. Chronic exposure to manganese may cause 'manganese psychosis'—an irreversible brain disease similar to Parkinson's disease. It is characterised by uncontrollable laughter, euphoria, impulsiveness, sexual excitement followed by impotence and speech disability.

Mercury. Mercury (Hg), known as quick silver, is a rare element, constituting less than 0.00008 per cent by weight of the earth's crust. It ranks 74th in the list of abundance elements in the earth's crust. Mercury is used in the manufacture of thermostats, discharge lamps, batteries, position sensors, relays and switches. In inorganic form, it attacks mainly liver and kidneys, whereas organo-mercurials traverse through the biological membranes and concentrate especially in brain creating erethism. Also being a teratogen, it is capable of inducing abortion and

embryotoxic effects.

Nonylphenol. Nonylphenol (NP) is a chemical, most widely known as a breakdown product of nonylphenol ethoxylate detergents, which is used as an antioxidant in some plastics and is found to cause damage to DNA and sperm function in humans.

Selenium. Selenium (Se) is an essential trace element that is extensively used in the manufacture of low-voltage rectifiers, photoelectric devices, photocopying equipment, etc. It is basically a non-metal with properties and reactions similar to sulphur. It falls below sulphur in the periodic table. Selenium is mutagenic and teratogenic and the central nervous system is its target. Generally, high toxicity of selenium is observed in humans, mainly due to the replacements of Thiol (-SH-) groups with SeH groups.

Polybrominated diphenyl ethers. Polybrominated diphenyl ethers (PBDE) are basically flame-retardants that are used to prevent the spread of fire in a wide variety of materials, including casings and components of many electrical and electronic products. This chemical is highly bio-accumulative and capable of interfering with brain development and hormones involved in growth and sexual development.

Triphenyl phosphate. Triphenyl phosphate (TPP) is one of the several organo-phosphorous flame retardants used in electronic equipment; for example, in the casing of computer monitors. It is acutely toxic and a strong inhibitor of a key enzyme system in human blood. It is also known to cause contact dermatitis and is a possible endocrine disruptor.

The Indian scenario

With rapidly changing domestic and global economic environment, the IT sector in India has undergone a sea change since early 1990s. In terms of production, internal consumption and export, electronics has emerged as the fastest growing industry. India's electronic equipment consumption is

estimated to reach \$126.7 billion this year and \$363 billion by 2015, growing annually at a rate of 29.8 per cent.

The problems associated with e-waste in India started surfacing after the first phase of LPG (liberalisation, privatisation and globalisation) policy during 1990s onwards. That decade witnessed a shift in economic policy, in turn, triggering off an increase in consumption pattern of the general public. These developments, along with indigenous technological advancements, led to addition of a wide gamut of e-waste churned out from households, commercial establishments, industries and public sector to the waste stream. The e-waste generated in India is expected to touch 470,000 tonnes by the year 2011.

India generated 330,000 tonnes of e-waste in 2009, while an additional 50,000 tonnes was imported illegally. Among the ten largest e-waste generating states, Maharashtra ranks first followed by Tamil Nadu, Andhra Pradesh, Uttar Pradesh, West Bengal, Delhi, Karnataka, Gujarat, Madhya Pradesh and Punjab. As of now, only 19,000 tonnes is processed in the country due to high cost involved and poor recycling infrastructure available.

There are just two formal recycling units located in the south and west of India and none in the northern and eastern India. While there is no large-scale recycling facility in the country, metropolitan cities like Delhi, Mumbai and Chennai have several spots where informal recycling of e-waste takes place.

In spite of the magnitude of the problem, there are no specific environmental laws or guidance related to e-waste management. None of the existing environmental laws has any direct reference to electronic waste or its handling or disposal as a hazardous material in nature.

Currently, in the absence of an effective mechanism, the obsolete electrical and electronic products in India meet with one of the following fates:

Equipment dismantling. Manual